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Remagnetization of the Rush Springs Formation, Cement, Oklahoma: Implications for dating hydrocarbon migration and aeromagnetic exploration

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ABSTRACT

The Permian Rush Springs Formation above the Cement anticline in Oklahoma contains a Late Permian–Early Triassic chemical remanent magnetization (CRM) that is interpreted to reside in authigenic magnetite. The CRM is found in bleached, carbonate-cemented sandstones that were altered by hydrocarbons and contain authigenic magnetite. The magnetite presumably precipitated in the Late Permian–Early Triassic as a result of chemical conditions created by hydrocarbons or associated fluids that migrated from underlying reservoir units. Red sandstones around Cement that were not altered by hydrocarbons contain a Permian CRM that resides in hematite. The red and bleached sandstones have similar magnetization intensities and susceptibilities; this raises questions about the use of aeromagnetic surveys in hydrocarbon exploration.

INTRODUCTION

Remagnetization and acquisition of a secondary magnetization by chemical processes are common phenomena observed in paleomagnetic studies of sedimentary rocks. The chemical remanent magnetizations (CRM) are acquired during precipitation of authigenic magnetic minerals or during replacement of precursor minerals by diagenetic magnetic phases. The CRM commonly reside in magnetite or hematite, and it is evident that several different chemical processes can cause remagnetization (e.g., Miller and Kent, 1988; Elmore et al., 1985). Paleomagnetic data, in conjunction with other studies, can provide important information on the origin and age of these remagnetizing events. One mechanism that has been documented as causing acquisition of a CRM in authigenic magnetite is related to hydrocarbon migration (Elmore et al., 1987). Paleomagnetic study of such a CRM therefore has potential for constraining the time of hydrocarbon migration.

In this paper we report results that constrain the time of hydrocarbon migration into the Rush Springs Formation at the Cement anticline in Oklahoma. The results also have implications for aeromagnetic hydrocarbon exploration, which has been suggested on the basis of studies of the Cement anticline (Donovan et al., 1979). The geologic setting at Cement also provides an opportunity to investigate directly, on a relatively small and manageable scale, how fluid (hydrocarbon) migration can cause remagnetization.

GEOLOGIC SETTING, PREVIOUS WORK AND METHODS

The Cement anticline, located in southwestern Oklahoma (lat 34° 57′N, long 98°12′W), is a buried Pennsylvanian structure that is unconformably overlain by a predominantly unfaulted Permian succession (Allen and Thomas, 1984). Subsurface structure at Cement is a northwest-southeast, doubly plunging, asymmetric anticline with two distinct domes, a boundary thrust fault on the north flank, and a normal fault system on the south side. The Permian succession is composed mostly of sandstone, siltstone, and shale, and the Rush Springs Formation (Guadalupian) occurs in the upper part of the succession. Minor structural deformation postdating the Rush Springs Formation produced a gentle upfold in the Permian units (Herrmann, 1961).

The Rush Springs Formation is typically a red, poorly cemented sandstone, but in a zone over the Cement anticline it is generally white to yellow and cemented with calcite and dolomite (Donovan, 1974). The facies in the unit were probably deposited in shallow-marine, shoreface, and coastal-plain environments under semiarid-to-arid climatic conditions (Allen and Thomas, 1984). Cement oil field has been an important source of oil and gas (Donovan, 1974).

On the basis of a study of the Rush Springs in the vicinity of the Cement anticline, Donovan (1974) suggested that subsequent to reddening, seepage of hydrocarbons from underlying reservoirs resulted in reduction and dissolution of hematite and bleaching of the sandstones above the anticline. Donovan et al. (1979) reported magnetite in near-surface bleached sandstones above the producing horizons at Cement oil field and suggested that hydrocarbons caused hematite to be altered to magnetite. They also suggested that this magnetite could be the source of an aeromagnetic anomaly observed over the oil field and that aeromagnetic surveys could be used in the exploration for hydrocarbons. Donovan (1974), as well as other workers (Lilburn and Al-Shaieb, 1984), also reported isotopically light carbonate cement at the crest of the anticline and attributed its presence to the oxidation of hydrocarbons as they migrated up from underlying reservoirs. These observations are the basis for a model consisting of red-bed formation followed by hydrocarbon migration, which caused dissolution of hematite, precipitation of magnetite and carbonate cement, and an increase in magnetic intensity.

The origins of magnetite and the magnetic anomaly associated with the Cement anticline are controversial. More recent studies of the magnetite reported by Donovan et al. (1979) indicate that it contains synthetic textures and is probably related to drilling activity (Reynolds et al., 1985). Reynolds et al. (1985) reported authigenic ferromagnetic pyrrhotite in well cuttings from the Cement field and suggested that the pyrrhotite precipitated as a result of hydrocarbon-related fluids. Boardman (1985), in a study of the aeromagnetic anomaly over the Cement oil field, suggested

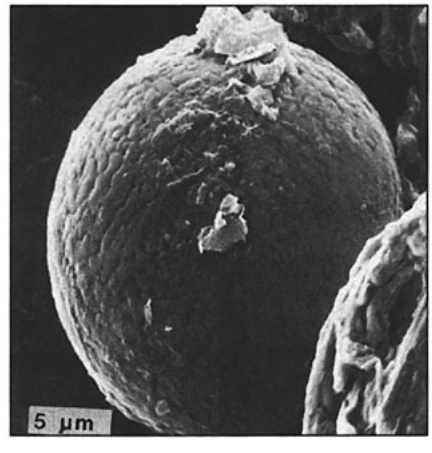


Figure 1. Scanning electron microscope photomicrograph of magnetic sphere extracted from bleached, carbonate-cemented sandstone at Cement that is interpreted to be authigenic magnetite.

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that the anomaly could be related to drilling equipment rather than diagenetic magnetite.

Although the presence of authigenic magnetite at Cement is open to question, other studies have documented a relation between hydrocarbons and magnetite authigenesis (e.g., Elmore et al., 1987; McCabe et al., 1987; Ellwood and Crick, 1988). Elmore et al. (1987) and Elmore and Crawford (1989) also presented evidence that chemical conditions created by hydrocarbons can cause acquisition of a CRM residing in authigenic magnetite.

Samples were collected from bleached sandstones at the Cement anticline (10 sites), red sandstones from the Rush Springs Formation 1.5 km from the anticline axis (one site), and reddish sandstones in an intermediate transition zone (seven sites). Samples were drilled with a portable gasoline-powered apparatus, and the cores were oriented with a Brunton compass and a clinometer. Magnetic measurements were made in a United Scientific three-axis cryogenic magnetometer located in a shielded laboratory. Both alternating field (AF) and thermal techniques were used to demagnetize the specimens. During AF demagnetization, specimens were subjected to 17 steps between the natural remanent magnetization (NRM) and 200 millitesla (mT). Specimens were thermally demagnetized in a stepwise manner (7–11 steps) using a Schonstedt TSD-1 unit. Direction and intensity results for each specimen were plotted as orthogonal projections, and best-fit lines were calculated for linear demagnetization trajectories by using principal component analysis (Kirschvink, 1980). Dips are shallow at the sampling localities (<2°), and no tilt correction was applied to the data. Isothermal remanent magnetization (IRM) acquisition and demagnetization experiments were also conducted for representative specimens. Magnetic susceptibilities of a representative set of specimens were measured using a Sapphire Instruments SI-2 magnetic susceptibility system and are reported in dimensionless International System (SI) volume units.

Thin sections of representative samples were analyzed under transmitted and reflected light. Some samples were also disaggregated and magnetically extracted. The extracts were examined by means of a scanning electron microscope (SEM) and energy dispersive analysis (EDA).

RESULTS

Bleached Sandstones

Carbonate-cemented bleached sandstones commonly have low porosities, and a petroleum odor is evident when the sandstones are freshly broken. The carbonate cement in many specimens contains hydrocarbon inclusions. Spheres (Fig. 1), in which iron is the only detected element based on EDA analysis, are found in the magnetic extracts. The spheres are similar to other spheres interpreted to be authigenic magnetite (e.g., McCabe et al., 1983).

Decay to 580 °C and the results of rock magnetic experiments (Fig. 2, A and B) suggest that magnetite carries the remanence in the carbonate-cemented bleached specimens. On the basis of rock magnetic experiments (Fig. 2B), pyrrhotite is probably also present. Mean magnetic susceptibility for carbonate-cemented bleached specimens is 2.8×10^{-5} SI (standard deviation = 7.8×10^{-6}, $n = 12$). Uncemented, bleached sandstones contain a weak ($<5 \times 10^{-5}$ amperes per metre [A/m]) and unstable magnetization.

Of the carbonate-cemented bleached specimens, approximately 55% contain a stable magnetization (commonly NRM ranges from 1 to 3×10^{-3} A/m) that decays in a north-northwest and shallow or south-southeast and shallow reversed direction during both AF and thermal demagnetization (Fig. 3, A and B; Fig. 4). Specimens with each direction are found in the same bed, and there is no stratigraphic control to the distribution of directions. In addition, the magnetization in some specimens with north-northwest directions moves toward the antipodal equivalent at high demagnetization steps. This suggests that the magnetization in these specimens is of dual polarity. The mean direction (Table 1) corresponds to a pole position at 56°N, 110°E (Fig. 5). When compared to the apparent polar wander path (APWP) of Irving and Irving (1982), the pole

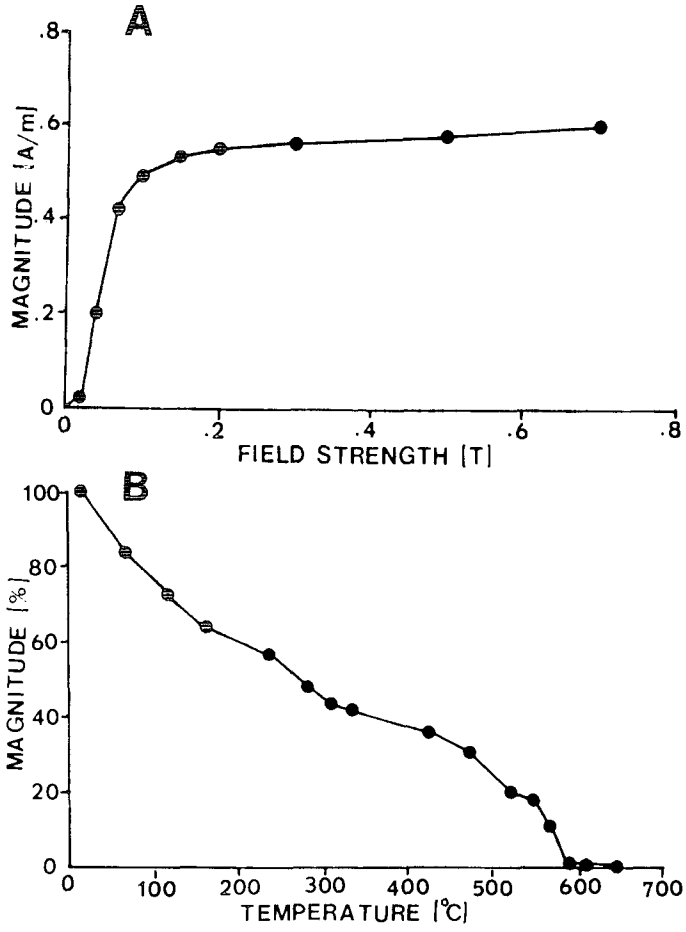


Figure 2. Representative isothermal remanent magnetization acquisition (A) and thermal decay (B) curves for bleached, carbonate-cemented specimens. Inflections in decay curve (B) at approximately 320 °C suggest presence of some pyrrhotite in samples.

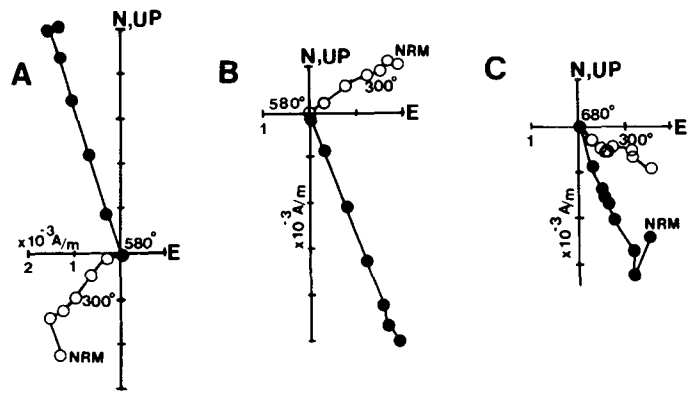


Figure 3. Representative orthogonal projections of thermal decay of natural remanent magnetizations from bleached, carbonate-cemented specimens (A and B) and red sandstone specimen collected 1.5 km from Cement (C). Specimens used in A and B were collected from same sample site, indicating no stratigraphic control. At high demagnetization temperatures, magnetic vector for specimen in A moves through origin toward south-southeast; this suggests magnetization may be of dual polarity. Solid symbols are horizontal components; open symbols are vertical components.

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position suggests remanence acquisition in Late Permian to Early Triassic time. This magnetization is interpreted to be a CRM on basis of the lack of stratigraphic control to the distribution of specimens with the normal and reversed directions, dual polarity magnetizations within individual specimens, and the presence of spheres interpreted to be authigenic magnetite. The presence of specimens with each direction in the same bed and the fact that the rocks were never deeply buried indicate that the magnetization is not thermoviscous in origin.

Most of the other carbonated-cemented, bleached specimens contain magnetizations that exhibit curved demagnetization trajectories, suggesting the presence of two or more unresolved components. Preliminary remagnetization circle analysis was not successful in resolving the components. A few of the other specimens contained "anomalous" directions (e.g., northeast and shallow).

TABLE 1. PALEOMAGNETIC DATA—CEMENT ANTICLINE

Rock Type	n	decl./incl. (°)	k	α95	Pole Position	A95
A	26	344/8*	16	7.4°	110°E 56°N	4.6°
B	9	165/9	74	6.0°	105°E 49°N	4.2°
C	34	11/39*	26	4.9°	40°E 75°N	4.4°

A= bleached, carbonate-cemented; B= red ss; and C= transition zone ss; n= number of specimens used in calculation; dec= declination; inc= inclination; k=precision parameter, α95= circle of 95% confidence, and A95= 95% confidence envelope around pole (Fisher, 1953). Pole positions calculated from sample pole positions.

*Reversed directions changed to antipodal equivalent for calculations.

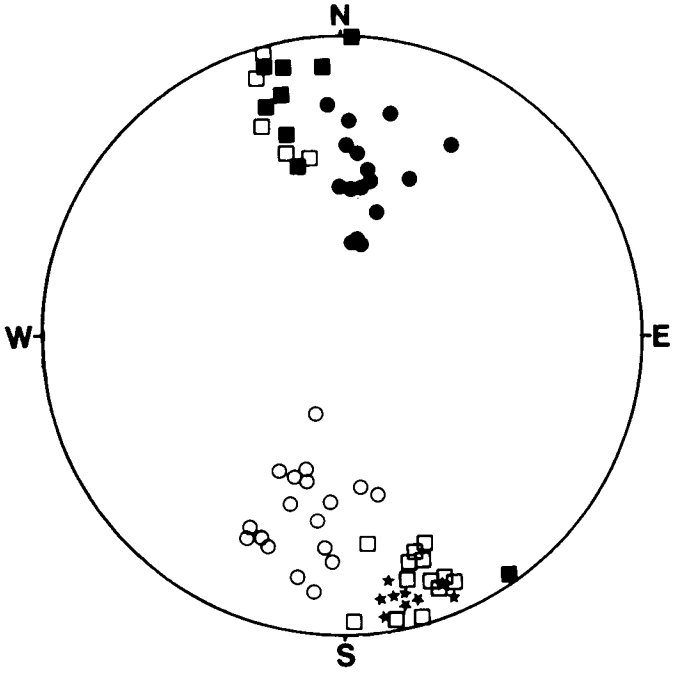


Figure 4. Equal-area projections of specimen directions from Rush Springs Formation. Solid symbols represent lower hemisphere projections. Open symbols represent upper hemisphere projections. Squares represent direction from bleached specimens. Stars represent red specimens collected 1.5 km from Cement. Circles represent red specimens collected from transition zone.

Red Sandstones

NRM intensities of both groups of red sandstones are about 1 to 3 × 10^{-3} A/m, and the mean magnetic susceptibility is 8.5 × 10^{-5} SI (standard deviation = 1.9 × 10^{-5}, n = 10). Linear decay up to 680 °C, and the results of IRM acquisition and decay experiments indicate that the magnetization in the sandstones is dominated by hematite. Petrographic studies indicate the presence of abundant authigenic hematite pigment in the red sandstones.

Specimens of sandstones collected 1.5 km from Cement contain a southeasterly and shallow direction (Fig. 3C; Fig. 4; Table 1). The pole (49°N, 105°E; Fig. 5) and the fact that all the directions are reversed suggest that the magnetization was acquired in the Permian during the 60 Ma Kiaman Reversed Superchron. The magnetic and petrographic characteristics of the specimens suggest that the magnetization is a CRM residing in authigenic hematite.

Specimen directions from the reddish sandstones in the transition zone, however, are either northerly and moderate down or southerly and moderate up (Fig. 4; Table 1). A pole position (Fig. 5; 75°N; 40°E) for the mean direction is off the APWP of Irving and Irving (1982) or a paleomagnetic Euler pole (PEP) path (e.g., May and Butler, 1986). The magnetization is probably a CRM, although the time of acquisition is problematic.

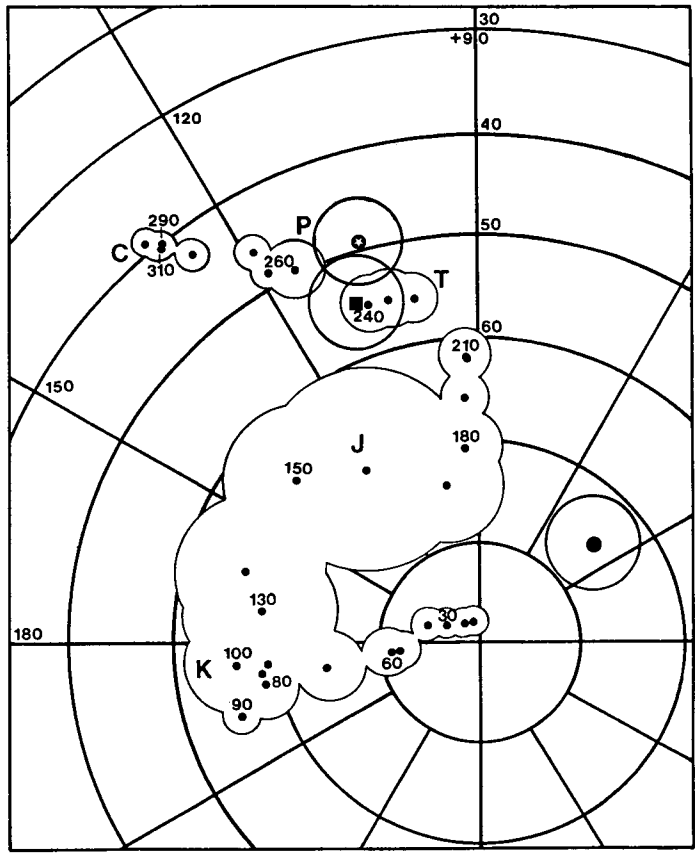


Figure 5. Apparent polar wander path from Irving and Irving (1982); square is paleopole for bleached specimens, star is paleopole for red sandstone away from Cement, and large circle is paleopole for red sandstones from transition zone. Poles are plotted with their envelopes of 95% confidence. Small dots are 10 Ma average pole positions (30 m.y. window) with circles of confidence corresponding to standard error. Letters correspond to age of path (C = Carboniferous; P = Permian; T = Triassic; J = Jurassic; K = Cretaceous).

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It could have been acquired in the Triassic-Jurassic, or it could represent a recent magnetization, offset from the axial geocentric dipole by secular variation.

DISCUSSION

The red and bleached sandstones have distinct magnetic and petrographic characteristics. A Late Permian–Early Triassic CRM interpreted to reside in authigenic magnetite is found in the bleached carbonate-cemented sandstones. The red sandstones away from the Cement anticline contain a Permian CRM residing in hematite. Sandstones in the transition zone contain a CRM residing in hematite, but the time of remanence acquisition is problematic. Also of interest are the observations that the bleached carbonate-cemented sandstones have NRM intensities that are similar to and magnetic susceptibilities that are somewhat lower than the red sandstones. These results have implications for constraining the time of hydrocarbon migration at Cement, for the origin of the magnetic anomaly at Cement, and for understanding some remagnetization mechanisms.

The results from the bleached, carbonate-cemented sandstones are consistent with the interpretation that authigenic magnetite formed because of the chemical conditions created by the hydrocarbons and/or associated fluids. The sandstones contain hydrocarbons and the precipitation of the carbonate cement was influenced by hydrocarbons. Pyrrhotite presumably also formed as a result of the migration of the fluids (Reynolds et al., 1985). If the hydrocarbons did cause precipitation of magnetite and acquisition of the CRM, then hydrocarbon migration occurred in the Late Permian–Early Triassic.

Although it is premature to suggest a specific geochemical mechanism for the precipitation of magnetite that is related to hydrocarbons, there are several possible sources for the iron needed to form magnetite, including iron in oil (Ellrich et al., 1985) and iron dissolved from the sandstones. It is also interesting to note that stable CRM residing in authigenic magnetite have only been reported from limestones (e.g., Elmore et al., 1987) or calcite-cemented sandstones. This suggests that carbonate may be a factor in controlling magnetite authigenesis.

The results from the red sandstones away from Cement suggest that remanence acquisition probably occurred before the end of the Kiaman Reversed Superchron in the Late Permian. These sandstones probably acquired their remanence as a result of hematite authigenesis during early diagenesis, as has been documented for some other red-bed units (e.g., Tauxe et al., 1987).

The Cement model (Donovan, 1974; Donovan et al., 1979) predicts an increase in magnetic intensity in the bleached sandstones compared to the red beds. This is not observed in our studies of surface samples. Although the precipitation of authigenic magnetite in the bleached, carbonate-cemented sandstones resulted in an increase in intensity, the hydrocarbons apparently caused hematite dissolution which reduced intensity. Studies of hydrocarbon-impregnated and bleached zones in other red beds (Kilgore and Elmore, 1989) also indicate that hydrocarbons do not cause an increase in intensity and, in fact, can cause a significant decrease. The intensity relations do not support the hypothesis that the magnetic anomaly at Cement is caused by the presence of magnetite and raise a general question about the use of aeromagnetic surveys in hydrocarbon exploration in red beds. The anomaly may be due to drilling equipment as suggested by Boardman (1985), although we cannot rule out the possibility that part or all of the anomaly is due to abundant authigenic magnetic phases in the subsurface.

The results from the bleached, carbonate-cemented sandstones suggest that the hydrocarbons caused precipitation of magnetite, which resulted in acquisition of a CRM and remagnetization. The hydrocarbons and/or associated fluids presumably migrated from underlying reservoir units along faults and fractures. Migration of orogenic fluids, including hydrocarbons, has been suggested as a process that can explain several geologic phenomena, including remagnetization (e.g., Oliver, 1986). The evidence for the proposed relation, however, is circumstantial. Although not exactly analogous to the geologic conditions thought to produce most orogenic fluids (e.g., Oliver, 1986), the results from the Cement anticline illustrate how fluids, which presumably migrated along conduits produced during deformation, can cause remagnetization.

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